

The Co-failure Möbius Conjecture

A Precise Statement for ICLR §4

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Re: UID 2016 Q3 (θ_{123} : precise statement, not a formalization-with-proof-strategy) and Q4 (§7 simulation confirmation). Statement stays entirely on the log-linear / probability side; cohomological home deferred to §8 and Clio-gated.

1. Setup

Fix an item population and three judges. For item x let $W_i(x) \in \{0, 1\}$ indicate judge i 's **failure** on x (co-failure = joint failure). Write the joint failure law P over $\{0, 1\}^3$ in its *log-linear (Ising / Möbius) form*:

$$\log P(w) = \sum_{S \subseteq \{1,2,3\}} \theta_S \prod_{i \in S} w_i \quad (\theta_\emptyset = \text{normalizer}).$$

Then $\{\theta_i\}$ are the singletons (marginal log-odds), $\{\theta_{ij}\}$ the pairwise couplings, and θ_{123} the single third-order coupling. Each θ_S is recovered by Möbius inversion over the subset lattice; in particular θ_{123} is the alternating-sign combination of the log joint cell-probabilities — a genuine function of P not reducible to lower-order marginals.

The conjecture below is stated in three parts because that structure maps directly onto **three falsifiable tests on FailureScope data**: genuineness (i) is a simple discrepancy test; edge-independence (ii) is a regression test on panel-level $(\theta_{123}, \theta_{ij})$ pairs; tail-consequence (iii) is a comparison of pairwise-predicted vs. observed all-fail rates (see Section 4 for details).

2. Conjecture (Co-failure Möbius / Three-body Coupling)

Only (i) is conjectural in the empirical sense; (ii) and (iii) are its algebraic consequences — (iii) is a theorem conditional on (i): once $\theta_{123} \neq 0$ is granted, the mis-estimation claim follows by algebra. The three-part structure is chosen because it separates what we conjecture about LLM judge panels from what is merely a consequence of that conjecture.

Conjecture 1 (Co-failure Möbius). *For LLM judge panels on a shared item population:*

(i) **Genuineness.** θ_{123} is generically nonzero. Equivalently: the pairwise-marginal-matched maximum-entropy model $\hat{P}_2$ (the Ising model fit to the three observed pairwise co-failure rates) does not reproduce the observed triple co-failure rate $P(W_1 = W_2 = W_3 = 1)$; the signed discrepancy equals θ_{123} to leading order.

(Empirically adjacent: Chen 2606.27288 reports a $2.5\text{--}3.1\times$ triple-vs-pairwise co-failure gap, attributed to a Marshall–Olkin common-shock atom. We do not re-claim that mechanism; θ_{123} is the log-linear invariant that quantifies the excess, whatever its mechanism.)

(ii) **Edge-independence.** θ_{123} is not determined by the edge data $\{\theta_{12}, \theta_{13}, \theta_{23}\}$: there exist panels with identical pairwise couplings and different θ_{123} . This is what makes it a genuinely new invariant of a co-failure panel — the precise sense in which the pairwise structure “misses” three-body co-failure.

(iii) **Tail consequence (the operational payoff).** A co-failure monitor whose null is the pairwise structure — equivalently, one calibrated on pairwise agreement / $\bar{\varphi}$ — mis-estimates the joint-tail (all-fail) probability by an amount controlled by θ_{123} , with sign:

$$\theta_{123} > 0 \implies \text{under-estimation of catastrophic joint failure.}$$

($\theta_{123} < 0$ gives over-estimation — the monitor is conservative; the safety-critical case is $\theta_{123} > 0$.) Consequently an effective sample size n_{eff} computed from pairwise $\bar{\varphi}$ alone is not tail-faithful when $\theta_{123} \neq 0$.

Remark (Algebraic home of θ_{123}). The value θ_{123} is the third-order Möbius coefficient of the joint failure law. Writing the log-density of the failure-indicator vector $(W_1, W_2, W_3) \in \{0, 1\}^3$ in its Möbius (log-linear / exponential-family) expansion,

$$\log p(w) = \theta_{\emptyset} + \sum_i \theta_i w_i + \sum_{i < j} \theta_{ij} w_i w_j + \theta_{123} w_1 w_2 w_3,$$

θ_{123} is the coefficient of the top monomial $w_1 w_2 w_3$. It is a real scalar, $\theta_{123} \in \mathbb{R}$ — the irreducible third-order (Lancaster–Streitberg) interaction of the joint Bernoulli. The ‘three-body coupling’ is precisely the part of the joint co-failure law not determined by the singletons $\{\theta_i\}$ and the pairwise couplings $\{\theta_{ij}\}$. Crucially, θ_{123} is Möbius-independent of the edge data $\{\theta_{ij}\}$ — it is not a function of the pairwise couplings — which is exactly what makes claim (ii) non-vacuous.

Whether θ_{123} additionally admits a cohomological reading, and if so whether it is a binary cup product or a higher (Massey-type) product on the nerve, is deferred to §8 (open, gated on Clio’s construction). We note only that, because θ_{123} is Möbius-independent of the edge data while a cup product of edge classes would be determined by that data, θ_{123} cannot be a simple cup product $a \cup b$ — consistent with $H^2(\Delta^2) = 0$ for the 2-simplex nerve.

3. Deferred Remark (Not Part of the Claim — Clio-Gated)

The cohomological home of θ_{123} is open. Edge-independence (ii) places it exactly in the regime where **cup products vanish and Massey-type products carry content**; whether θ_{123} is realized as a Massey product (not a binary cup product) is Clio’s question and must not be asserted in §4. §4 states (i)–(iii); §8 records the cohomological home as open.

4. Falsifiability

- (i) is refuted if $\hat{P}_2$ reproduces triple co-failure within sampling error across representative panels ($\theta_{123} \approx 0$).
- (ii) is refuted if θ_{123} is a deterministic function of $\{\theta_{ij}\}$ in the model class.
- (iii) is directly testable on **FailureScope**: compare the pairwise-predicted all-fail rate against the observed all-fail rate; the sign and magnitude of the gap is the leading-order θ_{123} .

This is why θ_{123} earns its place in §4 rather than sitting as an abstract curiosity: (iii) ties it to the monitor, which is the paper’s spine.

5. §7 Simulation Confirmation (Q4)

Adopting your call: **de Finetti two-atom mixture sweep** as the primary generative model for §7. Judges’ failures are conditionally independent given a latent Θ with a two-atom prior; sweeping the atom masses and separation drives the shared-latent correlation ρ from $0 \rightarrow 1$.

Power curve = detection vs. ρ , read off a quantity the reader already has from the setup. Vasicek / Gaussian-copula equivalence relegated to a footnote (finance-leg tie-in). One workshop-facing sentence included.

The naive-plug-in false-fire ($\sim 90\%$ under benign difficulty drift) and the paired e-process holding size are the (a)/(b) panels; (d) is the FailureScope application recovering $n_{\text{eff}} \approx 1.6$.